

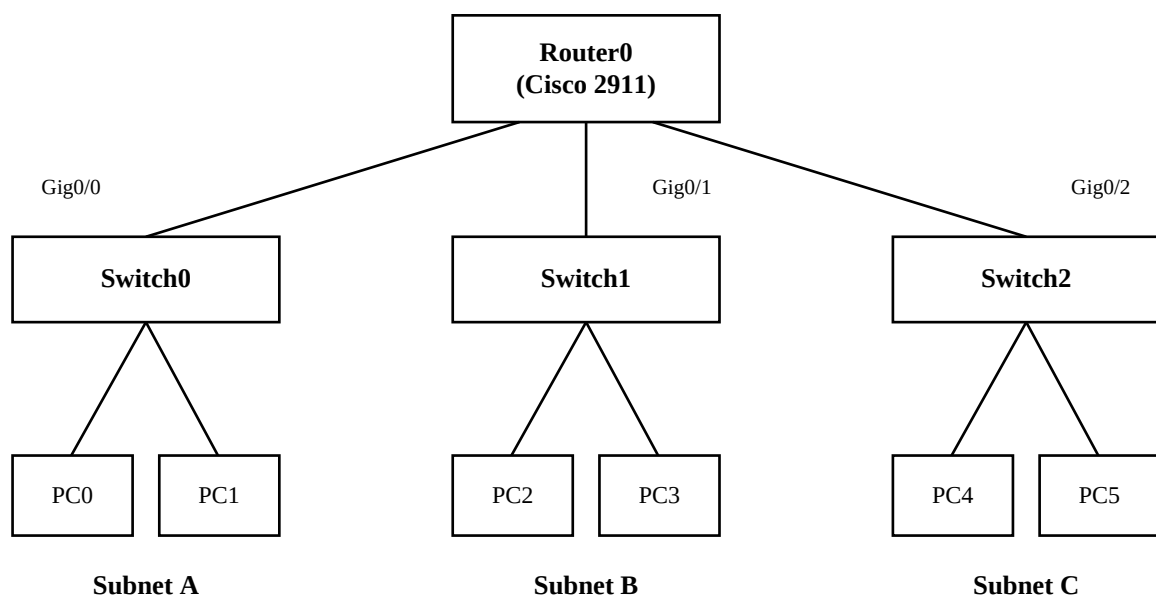
**Exp No: 9****Implementation of SUBNETTING in CISCO Packet Tracer Simulator****AIM:**

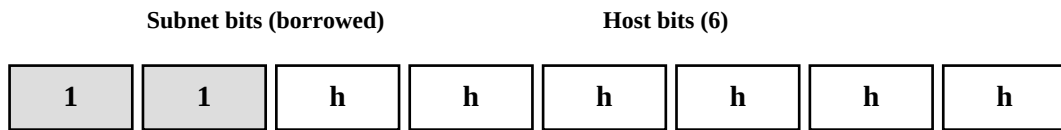
To design multiple subnets with a suitable number of hosts, assign static IP addresses, connect the subnets through a router in Cisco Packet Tracer, and observe packet transmission with and without the router.

**ALGORITHM / PROCEDURE:**

1. Take the network address 192.168.1.0/24 and decide the number of subnets needed (3 departments, so 4 subnets are created by borrowing 2 host bits).
2. The new mask is 255.255.255.192 (/26). Each subnet has 64 addresses, so 62 usable hosts.
3. Open Packet Tracer and place 1 router (2911), 3 switches (2960) and 6 PCs. Connect them with copper straight-through cables as in Fig 1.
4. Assign a static IP address, subnet mask and default gateway to every PC under *Desktop > IP Configuration*.
5. Configure the router interfaces: interface gig0/0, ip address 192.168.1.1 255.255.255.192, no shutdown
6. Repeat step 5 for Gig0/1 and Gig0/2 with the gateway address of subnets B and C.
7. From the PC command prompt use ping to test connectivity inside a subnet and across subnets.
8. Remove the router (delete the links to the router) and ping across the subnets again to see the difference.

**Fig 1: Network topology with three subnets joined by a router**



**Fig 2: Borrowing 2 bits of the host part for subnetting**

192.168.1.0 /24 -> /26 (2 bits borrowed = 4 subnets,  $2^6 - 2 = 62$  hosts each)

**Subnet design:**

| Subnet    | Network ID       | Mask            | Usable range | Broadcast     | Gateway       |
|-----------|------------------|-----------------|--------------|---------------|---------------|
| A (Sales) | 192.168.1.0/26   | 255.255.255.192 | .1 - .62     | 192.168.1.63  | 192.168.1.1   |
| B (HR)    | 192.168.1.64/26  | 255.255.255.192 | .65 - .126   | 192.168.1.127 | 192.168.1.65  |
| C (IT)    | 192.168.1.128/26 | 255.255.255.192 | .129 - .190  | 192.168.1.191 | 192.168.1.129 |
| Unused    | 192.168.1.192/26 | 255.255.255.192 | .193 - .254  | 192.168.1.255 | -             |

**IP address table:**

| Device  | Interface | IP Address    | Subnet Mask     | Default Gateway |
|---------|-----------|---------------|-----------------|-----------------|
| Router0 | Gig0/0    | 192.168.1.1   | 255.255.255.192 | -               |
| Router0 | Gig0/1    | 192.168.1.65  | 255.255.255.192 | -               |
| Router0 | Gig0/2    | 192.168.1.129 | 255.255.255.192 | -               |
| PC0     | Fa0       | 192.168.1.2   | 255.255.255.192 | 192.168.1.1     |
| PC1     | Fa0       | 192.168.1.3   | 255.255.255.192 | 192.168.1.1     |
| PC2     | Fa0       | 192.168.1.66  | 255.255.255.192 | 192.168.1.65    |
| PC3     | Fa0       | 192.168.1.67  | 255.255.255.192 | 192.168.1.65    |
| PC4     | Fa0       | 192.168.1.130 | 255.255.255.192 | 192.168.1.129   |
| PC5     | Fa0       | 192.168.1.131 | 255.255.255.192 | 192.168.1.129   |

**Router configuration:**

```

Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.1.129 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#end
Router#write memory

```

**OUTPUT:**

```

Router0 CLI - show ip interface brief

Router#show ip interface brief

```

| Interface          | IP-Address    | OK? | Method | Status | Protocol |
|--------------------|---------------|-----|--------|--------|----------|
| GigabitEthernet0/0 | 192.168.1.1   | YES | manual | up     | up       |
| GigabitEthernet0/1 | 192.168.1.65  | YES | manual | up     | up       |
| GigabitEthernet0/2 | 192.168.1.129 | YES | manual | up     | up       |

```

PC0 Command Prompt - same subnet

PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

```

**(a) When the subnets are connected via a router**

```

PC0 Command Prompt - ping PC2 (Subnet B) through the router

PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time=3ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)

```

```

PC0 Command Prompt - ping PC4 (Subnet C) through the router

PC>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=2ms TTL=127

```

PC0 Command Prompt - ping PC4 (Subnet C) through the router (contd.)

```

Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

```

### (b) When the subnets are not connected (router removed)

PC0 Command Prompt - ping PC2 (Subnet B) without the router

```

PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

```

PC0 Command Prompt - ping PC1 (same subnet) without the router

```

PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

```

### OBSERVATION:

| Case             | Source -> Destination           | Result                                                                        |
|------------------|---------------------------------|-------------------------------------------------------------------------------|
| Router connected | PC0 -> PC1 (same subnet)        | Success, TTL = 128                                                            |
| Router connected | PC0 -> PC2 / PC4 (other subnet) | Success, TTL = 127 (one router hop). First packet lost while ARP is resolved. |
| Router removed   | PC0 -> PC1 (same subnet)        | Success, switch forwards the frames                                           |
| Router removed   | PC0 -> PC2 / PC4 (other subnet) | Fails, 100% loss. No device can route between subnets.                        |

### RESULT:

Thus subnetting of the 192.168.1.0/24 network into /26 subnets was implemented in Cisco Packet Tracer. Hosts in different subnets could communicate only when a router was connected between them, while hosts inside the same subnet could communicate even without the router.